PCT/PCM 13: Score the Mask

4. Interpret the Figure

The right panel shows an all-background model achieving approximately 99.1% accuracy because most pixels belong to the background, even though it detects none of the foreground object. The zero Dice and sensitivity bars expose its failure: it has no overlap with the actual object and misses every foreground pixel.

5. Calculation

Given ground truth $|A| = 80$ pixels, prediction $|B| = 60$ pixels, and overlap $|A \cap B| = 48$ pixels:

$\text{Dice} = 2 \times 48 / (80 + 60) = 96 / 140 = 0.686$.
$\text{IoU} = 48 / (80 + 60 - 48) = 48 / 92 = 0.522$.

Verification: $\text{IoU} = \text{Dice} / (2 - \text{Dice}) = 0.686 / (2 - 0.686) \approx 0.522$. The relationship is verified.

6. Prediction Card

For an all-background model on an image with 2% foreground:

Accuracy = 98%.
Dice = 0.
Sensitivity = 0.

Accuracy is misleading because the model correctly labels the large background region but completely misses the foreground objects.